

410248 and 410256: Project Work Book (Guidelines and Log)

Fourth Year Computer Engineering

Year 2025-2026

Group/Project ID: 15

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Project Title: LegalSetu

Project Guide: Prof. Indu Jadhav

Area of the Project: Legal Compliance and Lawyer Consultation System



Department of Computer Engineering

Parvatibai Genba Moze College of Engineering

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Affiliated to Savitribai Phule Pune University**

Prologue

Project work is one of the most important components of the curriculum for the Engineering Graduates. From conceiving the idea to the materialization of it is a journey that has to be systematized, well defined and well documented to enjoy the full benefits of the efforts undertaken.

Every activity of the project development has its own importance and typical activities are like: Team formation, conceiving the idea, preparing the hypothesis, reporting the progress / development to the guide/ mentor, Interactions, suggestions and improvements, relevant documentations in proper format, schedule plans and visit logs.

Every institute is following their own best methods and techniques as per the guidelines and curriculum at the affiliated university. To bring the uniformity and standardization for the project work there is a need to come together and prepare the comprehensive guidelines regarding it.

This work book for the project work will serve the purpose and facilitate the job of students, guide and project coordinator. This document will reflect accountability, punctuality, technical writing ability and work flow of the work undertaken.

This document will definitely support the work undertaken.

Dr. Sachin Kharade
Coordinator, BoS Computer Engineering
SPPU, Pune

General Instructions

1. Students should enter the correct information in the work book.
2. Get all entries verified by respective project guide. No changes are to be made without project guide permission.
3. Students should report to their respective guides as per the schedule and its log is to be maintained in the work book.
4. Follow all deadlines and submit all documents strictly as per prescribed formats.
5. The work book should be produced at the time of all discussions, presentations and examinations.
6. The work book must be submitted to project coordinator/ guide/ department / College after successful examination at the end of year.
7. All documents and reports are to be prepared in Latex only (All the formats specifications provided adheres to MS Word but consequently applicable to final project report published using Latex)
8. Submit hard as well as soft copy. Maintain one copy with each member.

This booklet is supportive document to rules and a regulation provided by affiliated university curriculum providing recommendations, guidelines and is record of all related activities associated with project. This booklet is provided with the genuine intent to bring uniformity and to systematize the project work and to keep the audit of the work undergone by team members.

Work Book Development Project

Project Institution	Department of Computer Engineering, GSMT's Parvatibai Genba Moze College of Engineering, Pune
Support & Guidance	Dr. Somnath Kolgiri, Principal, GSMT's Parvatibai Genba Moze College of Engineering, Pune
Concept and Design	BoS Coordinator Computer Engineering , GSMT's Parvatibai Genba Moze College of Engineering, Pune
Project coordinator	Mrs. Poonam R , Assistant Professor, GSMT's Parvatibai Genba Moze College of Engineering, Pune
Technical Committee Members	1. Dr. Goraksh V. Garje 2. Dr. Parikshit Mahalle 3. Mr. Niranjan L. Bhale 4. Dr. Sunil R. Dhore 5. Dr. Nuzhat Saikh 6. Dr. Sudeep Thepade 7. Dr. Mangesh Bedekar 8. Ms. Swapnaja Hiray 9. Mr. Ranjeet Gawande
Date	17 th January 2025
Version No.	4.0
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Savitribai Phule Pune University, Pune

Computer Engineering

Program Educational Objectives

1. To prepare globally competent graduates having strong fundamentals, domain knowledge, updated with modern technology to provide the effective solutions for engineering problems.
2. To prepare the graduates to work as a committed professional with strong professional ethics and values, sense of responsibilities, understanding of legal, safety, health, societal, cultural and environmental issues.
3. To prepare committed and motivated graduates with research attitude, lifelong learning, investigative approach, and multidisciplinary thinking.
4. To prepare the graduates with strong managerial and communication skills to work effectively as individual as well as in teams.

Program Outcomes

Students are expected to know and be able -

1. To apply knowledge of mathematics, science, engineering fundamentals, problem solving skills, algorithmic analysis and mathematical modeling to the solution of complex engineering problems.
2. To analyze the problem by finding its domain and applying domain specific skills
3. To understand the design issues of the product/software and develop effective solutions with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4. To find solutions of complex problems by conducting investigations applying suitable techniques.
5. To adapt the usage of modern tools and recent software.
6. To contribute towards the society by understanding the impact of Engineering on global aspect.
7. To understand environment issues and design a sustainable system.
8. To understand and follow professional ethics.
9. To function effectively as an individual and as member or leader in diverse teams and interdisciplinary settings.
10. To demonstrate effective communication at various levels.
11. To apply the knowledge of Computer Engineering for development of projects, and its finance and management.
12. To keep in touch with current technologies and inculcate the practice of lifelong learning.

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1. About Project Work

The word *project* comes from the Latin word *project* from the Latin verb *proicere*, "to throw something forwards" which in turn comes from *pro-*, which denotes something that precedes the action of the next part of the word in time (paralleling the Greek *πρό*) and *iacere*, "to throw". The word "project" thus actually originally meant "something that comes before anything else happens".

(Curtsey Ref- <http://en.wikipedia.org/>)

The Project is conceiving the idea and implementing it systematically by using the knowledge derived in the course of education mainly to innovate or facilitate.

Work involves, study the feasibility of the project, planning project, studying existing systems, tools available to implement the project and state of art software testing procedures and technology with use of case tools.

The group of Under Graduate students at Final Year students will undertake project over the academic year. Work involves study the feasibility of the project, planning project, studying existing systems, tools available to implement the project and state of art software testing procedures and technology with use of case tools, design is to be implemented into a working model (software or hardware or both) with necessary software interface as an executable package.

a. Objectives and Outcomes:

Objectives -

- To apply the knowledge for solving realistic problem To develop problem solving ability
- To Organize, sustain and report on a substantial piece of team work over a period of several months
- To Evaluate alternative approaches, and justify the use of selected tools and methods,
- To Reflect upon the experience gained and lessons learned, To Consider relevant social, ethical and legal issues,
- To find information for yourself from appropriate sources such as manuals, books, research journals and from other sources, and in turn increase analytical skills.
- To Work in TEAM and learn professionalism.

Outcomes -

Students are expected to know and be able to-
Solve real life problems by applying knowledge.

Analyze alternative approaches, apply and use most appropriate one for feasible solution.

Write precise reports and technical documents in a nutshell.

Participate effectively in multi-disciplinary and heterogeneous teams exhibiting team work, Inter-personal relationships, conflict management and leadership quality.

b. Guidelines for Selection of Project Work:

Project is one of the significant contributory team works that has to be completed with distinct impression. It is really very difficult to explore the domain of interest / research/ thirst area/ society need. In Toto one cannot figuratively define best project but still there are certain parameters on which we can gauge the quality of project work done. It will be better suited to go for well-defined and relatively safe projects that provide scope for demonstrating proficiency with a low risk of failure especially at Under Graduate level.

General guidelines:

Identifying domain, feasibility and usability of work.

Project work is expected to involve a combination of sound background research (thorough study/ follow a line of investigation), and methodical implementation. Instead of fancied and driven behind the gaudy and ostentatious ideas, the utility has to be emphasized. It is also acceptable to identify the discrepancies/ flaws in the existing system and work accordingly to rectify or improve.

It is irrational to select the IDE and the software/ tools before the idea is not yet finalized.

Understanding the way project will be materialized and progressed.

c. Guidelines for Project Evaluation:

Project work is to be evaluated by both Internal and External examiners jointly, unanimously agreeing the following parameters among many others.

1. Problem definition and scope of the project
2. Through Literature Survey
3. Appropriate Software Engineering approach
4. Exhaustive and Rational Requirement Analysis
5. Comprehensive Implementation- Design, platform, coding, documentation
6. Optimization considerations (Memory, time, Resources, Costing)
7. Thorough Testing of all modules and integration of modules
8. Project Presentation and Demonstration (User Interface, ease of use, usability)
9. Presentation of work in the form of Project Report(s)
10. Understanding individual capacity, Role & involvement in the project
11. Team Work (Distribution of work, intra-team communication and togetherness)
12. Participation in various contests, Publications and IPR
13. Documents /Manuals (Project Report, Quick reference, System, Installation guide)
14. Outcomes / Usability / commercial value /product conversion of Work

(Refer Rubrics - page number 26)

2. University Syllabus

Term I

Teaching Scheme:

Tutorial: 2 Hours/Week

Term Work Assessment: 50 Marks

Course Objectives:

- To develop problem solving abilities using mathematics;
- To apply algorithmic strategies while solving problems;
- To develop time and space efficient algorithms;
- To develop software engineering documents and testing plans
- To use algorithmic solutions using distributed, Embedded, concurrent and parallel environments.
- To encourage and expose students for participation in National/ International paper presentation activities.
- Exposure to Learning and knowledge access techniques using Conferences, Journal papers and participation in research activities.

Course Outcomes:

To solve problem in projects;

To develop SRS and other software engineering documents in the project report;

To solve problems using multi-core, distributed, embedded, concurrent/Parallel environments;

To write conference paper;

To demonstrate presentation, communication and team-work skills.

Tools:

Preferably 64-bit FOSS tools but if sponsoring company's requirement is non-open source platform then it must be latest and current version of non-obsolete tools. 64-bit i5/i7/ Desktops/Mobiles, Latest SAN, 3-tier architectures along with latest version of FOSS Operating systems like Fedora 21 or equivalent, LAMP tools, Web server, Applications servers, Database servers, MongoDB or latest open source Big DATA tools, FOSS Programming Tools like gcc, g++, Eclipse, Python, Java and other tools are as per the requirement of the SRS. The documentation tools like Open office, GIT, Latex, Latex-Presentation.

Activity Planning for Tutorial Sessions:

I. Selection of Project Option and Framing the Problem to solve as a Project for the group of 3 to 4 students.

Option A: Industry Sponsored Project

Option B: Project as an Entrepreneur

Option C: Internal Project

II. Internal guide allocation for the BE Project:

Assistant Professor/Associate Professor/Professor as per AICTE norms in Computer Engineering having at least 5 years of full time approved experience can guide the BE

Project without compromising on the quality of the work (ref. Note1). The Project laboratory of 4 project groups (3 to 4 students in one group) constituting one laboratory tutorial batch (2 hrs per week), be allocated to the guide. The project group will submit the synopsis including title of the project, Technical Key Words (Ref. ACM Keywords) and relevant mathematics associated with the Project, names of at least two conferences, where papers can be published, Review of Conference/Journal papers (at least 10 papers + White papers or web references, (if any)) supporting the project idea, Plan of project execution using planner or alike project management tool. (Recommended dates: 3 weeks after Commencement of the Term). Preferably, the projects are Industry Sponsored or part of high level research/ Sponsored Research Project that are not conducted for any award of the educational degree or entrepreneurship project.

III. The project conduct and procedures are suggested as detailed below: -

Problem statement feasibility assessment using, satisfiability analysis and NP-Hard, NP-Complete or P type using modern algebra and relevant mathematical models. (recommended date of submission: - 8 weeks before term end).

IV. Use of above to identify objects, morphism, overloading, functions and functional relations and any other dependencies. (Recommended submission date: - 6 weeks before term end) Functional dependency graphs and relevant UML diagrams or other necessities. (recommended submission date: - 3 weeks before term end)

V. Testing of problem statement using generated test data (using mathematical models, Function testing principles) selection and appropriate use of testing tools, testing of UML diagram's reliability. (recommended submission date: - two weeks before term end)

VI. The index of submission must cover above mentioned 5 heads in addition to the instructions by the guide. Students must submit a Latex Report consisting of problem definition, literature survey, platform choice, SRS (System Requirement Specification) Document in specific format and high-level design document along with

Annex A: Laboratory assignments on Project Analysis of Algorithmic Design,

Annex B: Laboratory assignments on Project Quality and Reliability Testing of Project Design at the end of term-I and

Annex C: Project Planner and progress report after checking, removing/ avoiding the plagiarism.

Give an additional assignment per reporting plagiarism to be submitted in the report under the Annex heading extra-work. If the project is the replica of any other previous project or work from other unrelated persons than the student's team, such project should be rejected for the term work.

The term work at the end of Term-I shall be assessed and evaluated for 50 marks by the panel of examiners in the subject (Internal (preferably guide) and external examiner from Computer Department of Engineering Colleges). At-least one technical paper must be submitted on the project design in the conferences/workshops in IITs, Central Universities

or Uop Conferences or equivalent International Conferences Sponsored by IEEE/ACM and review comments received as Annex D. The examiners must seek answers regarding the suggestions given in the review comments of the paper submitted.

Term-I Project Laboratory Assignments: Tutorial Session

1. Refer Chapter 7 of first reference to develop the problem under consideration and justify feasibility using concepts of knowledge canvas and IDEA Matrix.
2. Project problem statement feasibility assessment using NP-Hard, NP-Complete or satisfiability issue using modern algebra and/or relevant mathematical models.
3. Use of divide and conquer strategies to exploit distributed/parallel/concurrent processing to identify objects, morphisms, overloading in functions (if any), and functional relations and any other dependencies (as per requirements).
4. Use of above to draw functional dependency graphs and relevant Software modeling methods, techniques including UML diagrams or other necessities using appropriate tools.
5. Testing of project problem statement using generated test data (using mathematical models, GUI, Function testing principles, if any) selection and appropriate use of testing tools, testing of UML diagram's reliability.

For Entrepreneurship type project additional assignments: Tutorial Session

6. To sign the MoU/agreement with the Engineering College for the Industry-on-Campus. The college shall provide the company the enclosure with lock-and-key to accommodate required table space, stabilized electricity and the Internet access. The College may host such company for first two years and further by renewing the MoU/Agreement. The college shall provide all the documents necessary for the establishment of the company. The College shall provide all the facilities as per agreement for Rent FREE, without any charges or fees or returns whatsoever for the First Year or Academic Duration of the activity. The college may prepare joint proposal with Industry for the AICTE/Government/University grants if any.
7. To study and establish a partnership company/proprietorship and get the PAN, MVAT, Profession Tax Number and such other necessary legal permissions.
8. Try and prepare clients list and communication with the clients or advertise the product by developing the Company WEB Site.
9. To submit Product Proposal for raising venture capital through government schemes of micro/small sector industries or through private venture capital entities.
10. To submit National/International patent/Copyright for first year to the Government Department of Patents and IPR.

Note 1. The guide for an entrepreneurship project shall be a full time approved Professor or Associate Professor possessing qualifications as per AICTE norms.

Note 2. If the students fail to complete the entrepreneurship assignments successfully then the project shall be treated as Internal Project for the purpose of assessment.

Note 3. All projects are expected to exploit multi-core, embedded and distributed computing wherever possible

Savitribai Phule Pune University

(Refer SPPU website for recent syllabus)

Semester II

Tutorials: 6 Hours/Week

Term Work: 100 Marks

Oral: 50 Marks

Course Objectives:

- To develop problem solving abilities using mathematics;
- To apply algorithmic strategies while solving problems;
- To develop time and space efficient algorithms;
- To develop software engineering documents and testing plans;
- To use algorithmic solutions using distributed, Embedded, concurrent and parallel environments.
- To encourage and expose students for participation in National/ International paper presentation activities.
- Exposure to Learning and knowledge access techniques using Conferences, Journal papers and participation in research activities.

Course Outcomes:

To solve problem and demonstrate the results of the project;
To develop SRS, reliability testing reports, and other software engineering documents in the project report;
To solve problems using multi-core, distributed, embedded, concurrent/Parallel environments;
To write conference paper;
To demonstrate presentation, communication and team-work skills.

Tools:

Preferably 64-bit FOSS tools but if sponsoring company's requirement is non-open source platform then it must be latest and current version of non-obsolete tools. Latest SAN, 3-tier architectures along with latest version of FOSS Operating systems like Fedora 21 or equivalent, LAMP tools, WEB server, Applications servers, Database servers, MongoDB or latest open source BigDATA tools, FOSS Programming Tools like gcc, g++, Eclipse, Python, Java and other tools are per requirement of the SRS. The documentation tools like Open office, Version control, Latex, Latex-Presentation.

1. Project workstation selection, installations and setup along with report to the guide. (Recommended submission date: - 3 weeks after commencement of second term)
2. Programming of the project, GUI (if any) as per 1st Term term-work submission. (recommended submission date: - Progress report every week during laboratory hours)
3. Test tool selection for various testing recommended by preferably external guide and generate various testing result charts, graphs etc. including reliability testing. (7 weeks before Term II Conclusion)
4. Review of design and necessary corrective actions taking into consideration feedback report of Term I assessment, and other competitions/conferences participated like IIT, Central Universities, University Conferences or equivalent centers of excellence etc.

5. Students must submit and preferably publish at least one technical paper in the conferences held by IITs, Central Universities or UoP Conference or International Conferences in Europe or US.
6. Final term work submissions in the prescribed format given by the guides consisting of a project report consisting of a preliminary report prepared in term-I, detailed design (all necessary UML diagrams) document, User Interface design, Laboratory assignments on test cases and test results generated by selected project testing tool, conclusions, appendix (if necessary), glossary, tools used and references at the end of Term-II after checking, removing/ avoiding the plagiarism. Give an additional assignment per reporting plagiarism to be submitted in the report under the Annex heading extra-work. If the project is the replica of any other previous project or work from other unrelated persons than the student's team, such project should be rejected for the term work.
7. The Term II examination is conducted by panel of examiners (preferably guide and expert from Industry having at least 5 years' subject experience or senior teacher in the subject in case of non- availability of industry expert). The project assessment shall be done using Live Project Demonstration (in existing functional condition), using necessary simulators (if required) and presentation by the students. The remarks of Term I assessment and related corrective actions must be assessed during examining the term-work.

Term-II Project Laboratory Assignments

1. Review of design and necessary corrective actions taking into consideration the feedback report of Term I assessment, and other competitions/conferences participated like IIT, Central Universities, University Conferences or equivalent centers of excellence etc.
2. Project workstation selection, installations along with setup and installation report preparations.
3. Programming of the project functions, interfaces and GUI (if any) as per 1st Term term-work submission using corrective actions recommended in Term-I assessment of Term-work.
4. Test tool selection and testing of various test cases for the project performed and generate various testing, result charts, graphs etc. including reliability testing.

Additional assignments for the Entrepreneurship Project:

5. Installations and Reliability Testing Reports at the client end.
6. To study Clients Feedback reports and related fix generations.
7. To create Documents will Profit and Loss accounts and balance-sheet of the company.

Note: If the students fail to complete the Entrepreneurship assignment successfully then the project shall be treated as Internal Project for the purpose of assessment.

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☎/ फ़ोन २५६५६२३४ Email : jdtepu@pn3.vsnl.net.in. Web.repune.org.in

दिनांक : 11 MAR 2009

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संज्ञ शिक्षण विभागीय कार्यालय, पुणे

4. Undertaking by Students

PARVATIBAI GENBA MOZE COLLEGE OF ENGINEERING PUNE

UNDERTAKING BY STUDENT

We, the students of B.E. Computer hereby assure that we will follow all the rules and regulations related to project activity for the academic year 2025-2026. The Project entitled-

“LegalSetu - Legal Compliance & Lawyer Consultation Portal”

will be fully designed/ developed by us and every part of the project will be original work and will not be copied/ purchased from any source.

Name of the student

Signature

1. ADIL SHAIKH

2. SHRIKANT GAVHALE

3. MOHIT BADGUJAR

4. Instructions Regarding Project Proposal and Finalization

1. The project work may involve the designing a system/subsystem or upgrading / improving an existing system. The design is to be implemented into a working model (software or hardware or both) with necessary software interface as an executable package (installable package or hardware model) along with
User & system manual and quick reference guide. A
project report including all necessary documents.
2. Group may come up with sponsored project. Sponsorship may not be in terms of money or resources. It might be in terms of just suggesting problem definition and associated guidance. Students may collect the letter required for applying the Institute/Industries for the project sponsorship from project coordinator
3. List of suggested projects, prominent domains and respective expert, whom you may contact for guidance, with Project Coordinator. Students may contact respective staff along with synopsis for the guidance. Students may contact respective staff for projects suggested by them in the respective areas.
4. Meet Project Coordinator for project title registration.
5. Synopsis must include project title, group members, sponsor details (if any), detailed problem definition, area, abstract, details of existing similar systems if any, scope of the project and software-hardware requirements. Sponsorship details include name of sponsoring authority, address, name of guide, sponsorship terms & conditions and respective documents certifying the same from authorities.
6. A Panel of experts will approve the project group and title only after presentation as per schedule. Presentation will cover details mentioned in the synopsis as above.

5. Schedule of Project Work Semester I

Sr.no	Activity Scheduled	Date
1.	Registration of Project groups	Mid of July
2.	Submission of Project Synopsis	Last Week of July
3.	Project presentations	First week of Aug
4.	Finalization of projects & allotment of guide	Second week of Aug
5.	Submission of final synopsis	Third week of Aug
6.	First presentation about progress of project work (Review I)	Last week of August
7.	Second presentation about progress of project work (Review II)	Second week of Sept
8.	Third Presentation (Review III)	Last week of Sept
9.	Fourth presentation about progress of project work (Review IV)	First week of Oct
10.	Submission of partial project report	Second Week of Oct
11.	Project work Examination	As per SPPU Notification

Semester II

Sr. No.	Activity Scheduled	Date
1.	Fifth presentation about progress of project work (Review V)	Mid of Jan
2.	Sixth presentation about progress of project work (Review VI)	Second week of Feb
3.	Review of Publication Activity	Last week of Feb
4.	Seventh presentation about progress of project work (Review VII)	Second week of March
5.	Submission of Draft of Report for checking	Third Week of March
6.	Submission of final project report and Project Work book to the project Coordinator	Last Week of March
7.	Mock Project Examination	First Week of April

8.	Project Examination	As per SPPU
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6. Copy of Proposal / Synopsis as per format (Annexure I)

	Genba Sopanrao Moze Trust's Paravatibai Genba Moze College of Engineering

Synopsis

- 1 **Group Id : Group No :-**
- 2 **Project Title :**
- 3 **Project Option :**
- 4 **Internal Guide :**
- 5 **Sponsorship and External Guide :**
- 6 **Technical Keywords (As per ACM Keywords)**

□

7 **Problem Statement**

8 **Abstract**

9 **Goals and Objectives**

Goal 1:

Objective 1.1:

Objective 1.2:

Goal 2:

Objective 2.1:

Objective 2.2:

Goal 3

Objective 3.1:

Objective 3.2:

10 Relevant mathematics associated with the Project

11 Software and hardware requirements for the implementation

Software Requirements:

and collaborating with others.

Hardware Requirements:

12 Innovativeness & Usefulness of Project

13. Names of Conferences / Journals where papers can be published

14. Review of Conference/Journal Papers supporting Project idea

15 Plan of Project Execution

Phase / Activity	Details / Tasks	Proposed Timeline	Team Member(s) Responsible
1. Problem Identification & Literature Survey	Define problem statement, collect requirements, review research papers & existing systems	Week 1 - Week 3 (Sem I)	All Members
2. Requirement Analysis	Hardware/software requirements, feasibility study	Week 4 - Week 5 (Sem I)	Member 1 & 2
3. System Design	System architecture, DFD/UML diagrams, module specifications	Week 6 - Week 8 (Sem I)	Member 2 & 3
4. Module Development - Phase I	Coding core modules / backend	Week 9 - Week 13 (Sem I)	Member 1 & 3

Phase / Activity	Details / Tasks	Proposed Timeline	Team Member(s) Responsible
5. Testing - Phase I	Unit testing of developed modules	Week 14 - Week 15 (Sem I)	Member 3
6. Project Progress Review (Sem I)	Documentation, partial implementation demo	Week 16 (End of Sem I)	All Members
7. Module Development - Phase II	Coding remaining modules, frontend, integration	Week 1 - Week 6 (Sem II)	Member 1 & 2
8. System Integration	Combine all modules, ensure compatibility	Week 7 - Week 8 (Sem II)	Member 2 & 3
9. Testing - Phase II	Functional testing, performance testing, debugging	Week 9 - Week 11 (Sem II)	Member 3
10. Final Deployment & Validation	Deploy on test environment, collect results, user feedback	Week 12 - Week 13 (Sem II)	All Members
11. Documentation & Report Preparation	Draft final report, IEEE paper (if applicable)	Week 14 (Sem II)	Member 3
12. Final Submission & Presentation	Final report, viva, demonstration	Week 15 - Week 16 (Sem II)	All Members

7. Project Review (Semester I)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented with thorough literature survey.

Project Review-I: Problem Statement, Motivation, objectives and

Literature Review

Student is expected to deliver presentation covering Problem Statement, Motivation, objectives and Literature Review.

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Does the statement gives clear identification about what your project will accomplish?	14/07/25		
2)	Is the statement short and concise?	14/07/25		
3)	Can a person who is not familiar with the project understand scope of the project by reading the project problem statement?	14/07/25		
4)	The project's objectives of study (what product, process, resource etc.) are being addressed?	18/08/25		
5)	Is similar type of methodology / model used for existing work?	25/08/25		
6)	Is the studied literature sufficient to decide scope of the project?	25/08/25		
7)	Are the objectives set will help to achieve goal of the project?	27/08/25		
8)	Does Research gap identified will lead to find motivation of project?	29/08/25		
9)	Does your project contribute to our society by any means and will lead to find motivation?	31/08/25		
10)	Are the objectives clearly and unambiguously listed?	31/08/25		

Remark and Suggestions:

Name and Sign of Reviewers:

1.

2.

Project Review-II: Feasibility and Scope

Student is expected to deliver presentation covering Feasibility and Scope

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Is the project's view point is understood?	20/08/25		
2)	Is the project goal statement is in alignment with the sponsoring organization's business goal and mission?	22/08/25		
3)	Who is the project's end user?	25/08/25		
4)	What is the projected cost of producing a product?	27/08/25		
5)	Is project achievable in specified (Time, Cost Budget)?	29/08/25		
6)	Are the requirements within the scope of the project?	01/09/25		
7)	Is the scope properly defined?	03/09/25		
8)	Does the problem statement clearly define scope of the project?	05/09/25		
9)	Do the project requirements fit into available software and hardware?	08/09/25		
10)	Whether the milestones are stated completely and project timeline is given?	10/09/25		
11)	Whether risks like technical risks, Operational risks, schedule risks, Business risks are identified correctly or not?	12/09/25		
12)	Whether Risk prioritization is done properly or not and any back up plan is there or not?	14/09/25		
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

Project Review-III: Requirement Analysis

Student is expected to deliver presentation covering Requirement Analysis

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Is information domain analysis complete, consistent and accurate?	10/10/2025		
2)	Is problem statement categorized in identified area and targeted towards specific area there in?	12/10/2025		
3)	Is external and internal interfacing properly defined?	12/10/2025		
4)	Are requirement consistent with schedule, resources and budget?	12/10/2025		
5)	Are all requirements traceable to system level?	12/10/2025		
6)	What is needed to make the product?	12/10/2025		
7)	Is there a demand for the produce?	12/10/2025		
8)	Is identification of stakeholders is done properly?	12/10/2025		
9)	Whether all requirements are captured and documented in line with scope?	12/10/2025		
10)	Whether all type of analysis classes are identified or not?	12/10/2025		
11)	Whether the Acceptance criteria is decided are not?	12/10/2025		
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

Project Review-IV: Design

Student is expected to deliver presentation covering Design

Sr. No.	Question	Date	Remark / Grade	Sign of Guide
1)	Are requirement reflected in the system architecture?	10/10/2025		
2)	Does the design support both project (product) and project goals?	12/10/2025		
3)	Does the design address all the issues form the requirement?	14/10/2025		
4)	Is effective modularity achieved and modules are functionally independent?	16/10/2025		
5)	Are structural diagrams (class, Object, etc.) are well defined?	18/10/2025		
6)	Are all class associations clearly defined and understood?(Is it cleat which classes provide which services)?	20/10/2025		
7)	Are the classes in the class diagram clear? (What they represent in the architecture design document?)	22/10/2025		
8)	Is inheritance appropriately used?	24/10/2025		
9)	Are the multiplicities in the use case diagram depicted in the class diagram?	26/10/2025		
10)	Are all objects used in sequence diagram?	28/10/2025		
11)	Are the symbols used in all diagrams corresponding to UML standards?	30/10/2025		
12)	Are behavioral diagrams (use case, sequence, activity, etc.) well defined and understood?	31/10/2025		
13)	Does each case have clearly defined actors and input/ output?	31/10/2025		
14)	Does the sequence diagram matches with class diagram?	31/10/2025		
15)	Is aggregation/ containment (used) clearly defined and understood?	31/10/2025		
16)	Whether State charts are capturing system's dynamic behavior correctly or not?	31/10/2025		
17)	Related to procedural thinking whether DFDs and CFDs along with transaction and transformation flow are done correctly or not?	31/10/2025		
Remark and Suggestions:				

Name and Sign of Reviewers:

- 1.
- 2.

8. Internal Evaluation Sheet (Semester I)

Sr. No.	Name(s) of the student in the project group	Problem Statement / Motivation / Objectives / Scope/ Feasibility Requirement (05)	Literature Survey (05)	Requirement Analysis (05), Modeling & Designing (10)	Planning & Prototyping (05)	Presentation & Question - Answer (10)	Partial Project Report (10)	Total (50)
1.								
2.								
3.								
4.								

(Refer Rubrics - page number 26)

Name and Signature of Evaluation Committee:

1. Prof.

2. Prof.

Examiners Feedback and Suggestions:

Signature of Guide
[]

Signature of Head

Head of Department

9. Project Review: (Semester II)

The group members are expected to present their work undertaken during the semester. Journey of development has to be rationally presented.

Project Review-I: Modeling (Model Refinement and Algorithm development)

Student is expected to deliver presentation covering Modeling

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Which software Development Process model is used? (Water fall, Incremental, RAD) How?(? at this level?)	04/01/2026		
2)	Do you clearly identify data objects, their attributes and relationships?(All constraints fro SRS are captured or not?)	04/01/2026		
3)	Have you clearly matched the objects with respective classes and their responsibilities?	05/01/2026		
4)	Have you analyzed the requirements and represented them into respective models?	05/01/2026		
5)	Can you differentiate between different system states and depict them in the form of state transition diagram?	06/01/2026		
6)	Does the mathematical model clearly imply design of the project?	07/01/2026		
7)	Does the mathematical model clearly states goal of project?	08/01/2026		
8)	Does the interface between the modules properly identified?	14/01/2026		
9)	Does any functional dependencies are identified and described?	15/01/2026		
10)	Which architectural model does your system supports?	19/01/2026		
11)	Whether Deployment diagram is inline with selected architecture?	19/01/2026		
12)	Whether all components are designed properly and represented in component diagram?	23/01/2026		
13)	Whether NP-completeness of algorithms is checked or not?	25/01/2026		
Remark and Suggestions:				

Name and Sign of Reviewers:

- 1.
- 2.

Project Review-II: Coding / Implementation

Student is expected to deliver presentation covering Coding / Implementation

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Does the code completely and correctly implement the design?	04/02/2026		
2)	Does the code comply with the coding standard?	04/02/2026		
3)	Is the code well structured, consistent in style, and consistently formatted?			
4)	Are all functions in the design coded?	05/02/2026		
5)	Does the code make use of object oriented concepts?			
6)	Does the code support granularity?	05/02/2026		
7)	Does the language used for coding is correctly chosen as per the project need?	06/02/2026		
8)	If any off the shelf components are used, Have you understood the functionalities of using it?	07/02/2026		
9)	Are all comments consistent with the code?	08/ 02/2026		
10)	Whether code optimization is done properly or not?(By using language features)			
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

Project Review-III: Validation and Testing

Student is expected to deliver presentation covering Validation and Testing

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Have you done alpha testing?	14/03/2026		
2)	Have you done beta testing?	14/03/2026		
3)	Have you validated the requirements, design and code as per standard?	14/03/2026		
4)	Have you performed GUI testing of project? How?			
5)	Does your system comply with basic usability norms?	15/03/2026		
6)	Have you tested the code using standard datasets available in your area of project?	15/03/2026		
7)	Have you tested the code in real time environment?	16/03/2026		
8)	After integration of all components whether total performance of system is checked or not?	17/03/2026		
9)	Whether repository of all components along with versions is documented or not?			

Remark and Suggestions:

Name and Sign of Reviewers:

1.

2.

Project Review-III: Report Writing

Student is expected to deliver presentation covering Report Writing

Sr. No.	Question	Date	Remark/ Grade	Sign of Guide
1)	Is the report written as per the prescribed format?	24/04/2026		
2)	Is the report timely prepared?	24/04/2026		
3)	Is the report properly organized, spelled, grammatically correct?	24/04/2026		
4)	Is the report plagiarism free?	25/04/2026		
5)	Is the report precise and written to the point?	25/04/2026		
6)	Is the report contains complete results and comparative graphs?	25/04/2026		
7)	Are all figures and tables properly numbered and labeled?	26/04/2026		
8)	Are all figures and tables properly cited?	26/04/2026		
9)	Weather references are properly cited?	27/01/2026		
Remark and Suggestions:				

Name and Sign of Reviewers:

1.

2.

10. Internal Evaluation Sheet (Semester II)

Sr. No.	Name(s) of the student in the project group	Modeling (10)	Coding And Implem-Entation (40)	Testing (10)	Understa nding, Individual Involvem-ent / Contribut-ion in the project (10)	Team Work (10)	Demons-tration cum Present-ation (10)	Docu-ment s & Repo rt (10)	Total (100)
1.									
2.									
3.									
4.									

(Refer Rubrics - page number 26)

Name and Signature of Evaluation Committee:

1. Prof.

2. Prof.

Examiners Feedback and Suggestions:

Signature of Guide
[]

Signature of Head

Head of Department

11. Software Engineering Code of Ethics and Professional Practices

(Courtesy / Reference- <http://www.acm.org/about/code-of-ethics>)

Computers have a central and growing role in commerce, industry, government, medicine, education, entertainment and society at large. Software engineers are those who contribute by direct participation or by teaching, to the analysis, specification, design, development, certification, maintenance and testing of software systems. Because of their roles in developing software systems, software engineers have significant opportunities to do good or cause harm, to enable others to do good or cause harm, or to influence others to do good or cause harm. To ensure, as much as possible, that their efforts will be used for good, software engineers must commit themselves to making software engineering a beneficial and respected profession. In accordance with that commitment, software engineers shall adhere to the following Code of Ethics and Professional Practice.

The Code contains eight Principles related to the behavior of and decisions made by professional software engineers, including practitioners, educators, managers, supervisors and policy makers, as well as trainees and students of the profession. The Principles identify the ethically responsible relationships in which individuals, groups, and organizations participate and the primary obligations within these relationships. The Clauses of each Principle are illustrations of some of the obligations included in these relationships. These obligations are founded in the software engineer's humanity, in special care owed to people affected by the work of software engineers, and the unique elements of the practice of software engineering. The Code prescribes these as obligations of anyone claiming to be or aspiring to be a software engineer.

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following Eight Principles:

1. PUBLIC - Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER - Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
4. JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.
5. MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues.
8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Environment and Computing-

Information and communication technologies (ICTs) have been contributing to environmental problems: computers, electronic devices and ICT infrastructure consume significant amounts of electricity, placing a heavy burden on our electric grids and

contributing to greenhouse gas emissions. In 2007, the total footprint of the ICT sector - including personal computers (PCs) and peripherals, telecoms networks and devices and data centers - was 830 Mt CO₂ emission, about 2% of the estimated total emissions from human activity released that year (a figure equivalent to aviation). ICT hardware poses severe environmental problems both during its production and its disposal. Each stage of a computer's life, from its production, throughout its use, and into its disposal, presents environmental problems. Manufacturing computers and their various electronic and non - electronic components consumes electricity, raw materials, chemicals, and water, and generates hazardous waste. All these directly or indirectly increase carbon dioxide emissions and impact the environment and the trend is to increase in the BAU (Business As Usual) scenario.

Green Computing-

Hence you all our students are requested to follow green computing practices. Green computing is the study and practice of designing, manufacturing, using, and disposing of computers, servers, and associated subsystems – such as monitors, printers, storage devices, and networking and communications systems –efficiently and effectively with minimal or no impact on the environment. Green computing includes the dimensions of environmental sustainability, the economics of energy efficiency, and the total cost of ownership, which includes the cost of disposal and recycling. Green computing benefits the environment by improving energy efficiency, lowering greenhouse gas emissions, using less harmful materials, and encouraging reuse and recycling. Green design, Green manufacturing, Green use, Green disposal are complementary paths of green ICT. Only focusing on these four fronts we can achieve total environmental sustainability from the IT side and make IT greener throughout its entire lifecycle.

Social Life and Computing-

Each IT professional must keep in mind the three key components of a corporate Green IT best practices policy -Environment, Economy and Social aspect. The invention of the computer has completely changed the way we live our lives. Nearly everything is controlled by a computer; cars, satellites, phones, etc. Computers have made our lives easier. Computers can also have positive effects on a person's social life when their power to connect over great distances is harnessed fully. Computers have both positive and negative impact in our society. While technology is a wonderful thing it is almost likely that it can be used in an immoral or wrong way. There is a price to pay for everything even if it appears it's making life easier on people.

While proper lifecycle management can greatly boost a IT company's ecological and environmental sustainability position, it can also contribute to achieving goals on the social front. Hardware retirement practices are the primary concern in this regard. In addition to seeking carbon neutrality, a proper asset retirement strategy should seek sustainability in the communities where companies operate.

The following social objectives should be considered:

1. To optimize sustainability in their IT infrastructure, companies should focus on each state of the IT lifecycle
2. Setting the Appropriate Corporate Sustainability Policy
3. Avoiding unethical labor practices and Controlling unethical exports
4. Accountability in the Recycling e-waste and Sustainability Metrics and Reporting
5. Greater Transparency Regarding Material Analysis and Extraction
6. Compliance with stringent, evolving security regulations

12. Contest Participation Details.

A. Participation in project Competition / Contest

Sr. No.	Name and Place of Project Competition and Exhibition	Date	Certificates prizes won if any
1.			
2.			
3.			
4.			

Attach attested copy of certificate(s)

B. Paper Publication/ Presentation/IPR

Sr. No.	Name of Organizer	Date	Certificates/ Prizes won if any
1.			
2.			
3.			
4.			

Attach attested copy of certificate(s)

13. Rubrics

A. Idea Inception

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Problem Definition and Scope of the Project				
Literature Survey				
Software Engineering Approach				
Requirement Analysis				

B. Implementation

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Implementation- Design, platform, coding,				
Optimization considerations(Memory, time, Resources, Costing)				
Thorough Testing of all modules				
Integration of modules and project as whole				

C. Documents

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Synopsis				
Project Report				
Quick references				
System manual				
Installation Guide				
Work Book				

D. Demonstration

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Project Presentation and Demonstration(User Interface, ease of use, usability)				
Understanding individual capacity & involvement in the project				
Team Work (Distribution of work, intra-team communication and togetherness)				
Outcomes / Usability				

E. Contest Participation / Awards, Publications and IPR

Grade (Grade Point)	Excellent (10-9)	Very Good (6-8)	Fair (3-5)	Poor (1-2)
Parameter				
Participation in various contests				
Appreciation and Awards				
Publications				
Copyright				
Patent				
Commercial value /product conversion of Work				

Bibliography

1. Roger S. Pressman, "Software Engineering: A Practitioner's Approach", 6th Edition McGraw- Hill, ISBN978-0-07-337597-7.
2. Joseph Phillips, "IT Project Management", Tata McGraw-Hill 2003 Edition, ISBN 13: 978-0071700436
3. www.csc.villanova.edu/~tway/courses/csc4181/s2010/srs_template-1.doc
4. http://unipune.ac.in/Syllabi_PDF/revised-2015/engineering/BE-Computer-2012-course-27-8-15.pdf

Annexure i: Final Synopsis (after approval of the project work)

Title Page

Project Group ID/ group

Details Title of the project

Domain such as databases, image processing, network based, web technology based etc.

Team Members (List with Signatures)

Sponsorship details if any (Name, External Guide name and Designation with Signature, e- Mail ID)

Internal Guide (with signature of approval)

Inner Pages:

Keywords (ACM Keywords)

Problem Definition

List of modules/ functionalities

Current market survey: This should include list of similar products available, if any and also their pros and cons.

Scope of the project

Literature survey (List of references only): This should include the list of books, magazines, research papers, web links etc. referred by the students.

Software and hardware requirements of the project

Probable date of completion

Outcomes

Annexure ii: Partial Project Report (Semester I)

A preliminary report of project work (Partial Project Report) is to be prepared as per the guideline given below using **Latex** and is to be submitted at the end of semester I.

Title Page

First page containing Name, Topic Name, Guide Name, Year, Branch, and College Name etc. (see format displayed herewith)

Certificate

Certificate (provided by college)

Project approval sheet (see format displayed herewith)

Certificate certifying the project work done approved by the sponsoring authority, if any.

Acknowledgements (if any)

Thanking any person / staff member / friend if to be done so.

Abstract

A minimum of 100 words briefing the topic in consideration.

Keywords

A minimum of 5 and maximum of 10

Introduction

Introduction should be minimum of 200 words, briefing of the details to follow. It should cover details of project work, objectives, scope of the project and organization of report.

Literature Survey

The purpose of the literature survey is to identify information relevant to project work and the potential and known impacts of it within the project area. This section should include a comprehensive report of current market survey done with respect to problem. Include study of similar systems available, if any along with their pros and cons. identify those areas where there is an absence or scarcity.

Design Details (Phase I to IV)

Phase I: Requirements Analysis

The Group is to submit a detailed write - up indication the requirements that the project demands, viz.

Actual detailed problem definition.

The definition is to include all that is to be done and is to be put up in the final software and / or

Hardware (product) that is to be generated from the years' work (User's point of view).

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document.

Modified Requirements (After doing feasibility study) are to be prepared under all the 3 categories listed above from the developer's point of view. The requirement listed herein should be feasible technically from the software / Hardware point of view. The new list is also be categorized in the 3 categories listed above.

Follow the standard format of SRS.

Phase II: Analysis Phase

The group (based on Phase I) is to suggest the paradigm followed by them in the project. The paradigm should be justifiable from Phase I. The various stages and work to be completed under them is to be indicted in detail.

Phase III: Design Phase

ERDs (Optional, decide in consultation with guide)

The group is to draw the ERD (Entity Relationship Diagram) for the project. (This should be justifiable with regard to Phase I & II)

The ERD after getting evaluated (by dry running) is to be analyzed for incompleteness from any point of view.

The ERD thus validated should be made fair in a presentable fashion.

This ERD is to be included in the Report.

IF

The project group is to follow an “Object Oriented” Approach for their Project.

THEN

The group will draw all (all the nine) UML (Unified Modeling Language) diagrams for the project.

These diagrams are to be refined in every aspect for this report (as per requirements finalized in phase I)

Proper notations are to be used in all the figures drawn.

Proper Color-coding if required is to be used.

Extensions to diagrams / customizations may be done and represented (if the project demands it)

ELSE (groups following Structured Approach)

The group will draw the DFD-s (Data Flow Diagrams) for the Project. (These should be justifiable with respect to Phase I, II and the ERD)

DFD Level 0, Level 1, Level 2 should be drawn in an evolutionary fashion (No entries to appear in Level 2 unless they are in Level 1, which in turn are in Level 0)

The DFD's are to be validated and made final in a presentable fashion. Proper Color- coding is expected

Extensions to DFD-s may be represented (if the project demands it)

Phase - IV: Planning Phase

The group is to finalize the Front End/ Back End required for the project as per the demands of the project (Software and / or hardware)

The Front End/ Back End should be justifiable depending on the complexity of the project.

The structure of the database to be finalized depending on the complexity of the project.

Any Normalization required on the database is done so as to ensure correctness for the future phase.

Coding Language / Methodology should be finalized/

Time requirement to be finalized and indicated

Actual project plan including major milestones should be decided and finalized

Rough estimates of lines of code / functions / routines to be made.

Rough estimates of lines of code / Objects / Classes to be made (for Groups following OO Paradigm)

Software Reuse / Re - Engineering possibilities are to be expected and indicated
Software and Hardware requirement.

Probable date of completion.

Scope of the project.

Phase - V: Prototyping

A prototype is expected which basically includes all the MAJOR features in the project.

The GUI/ Front end is to be prepared.

The structure of the database / back end (if any) to be indicated.

The prototype is built basically to give a feel of the actual software and / or hardware (Product) that is expected

Major routines / Functions are expected.

Conclusions

Include conclusions from the work done with a minimum of 50 words

References

List out Books, Magazines, Thesis, Journals, Web links etc. referred in IEEE format

Plagiarism Check Report

Format of SRS

Software requirement Specification is a detailed write-up indicating the requirements that the project demands. It contains actual detailed problem definition. The definition is to include all that is to be done and is to be developed in the final software and / or Hardware (product) that is to be generated from the years' work (User's point of view). The entries under this section are to be categorized under the categories,

1. Necessary functions,
2. Desirable functions, and others

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document. Modified requirements (after doing feasibility study) are to be prepared under all the 3 categories listed above from the developer's point of view. The requirements listed herein should be feasible technically from the software/ Hardware point of view.

It should include following important requirements.

1. Detailed Problem Definition

2. External Interface Requirements

User interfaces
Hardware Interfaces
Software Interfaces
Communication Interfaces

3. System Features

Feature 1
Feature 2 etc.

4. Other Non- functional requirements.

Performance requirements
Safety requirements
Software Quality attributes

Annexure iii: Project Report (Semester II)

A report of project work is to be prepared as per the guidelines given below using Latex and is to be submitted at the end of semester II along with CD containing (copy of Partial Project Report, Final Project Report along with .tex files, Power point presentation, copy of base paper and reference papers, executable Project Code, supportive software platform for the project execution).

First page as per standard college reports

First page containing Name, Topic Name, Guide Name, Year, Branch, and College Name etc. (see format displayed herewith)

Certificate (will be provided by college)

Dissertation approval sheet (see format displayed herewith) Also attach certificate certifying the project work done approved by the sponsoring authority, if any.

Abstract

A minimum of 100 words briefing the topic in consideration.

Keywords

A minimum of 5 and maximum of 10

Index

Details of various Topics, Sub-Topics, with Page No. Figure Index, giving details of page number, figure number and figure caption Table Index, giving details of page number, table number and table caption (If any) Index of Pseudo-code / Sample code (If any)

I) Introduction

Minimum of 200 words, giving some briefing of the details to follow.

- Detailed problem definition
- Justification of problem
- Need for the new system
- Advances/additions/updating the previous system
- Presently available systems for the same
- Purpose of your system
- Organization of the report

This section should be relevant to the Literature Survey done and reported in the partial project report. The purpose of the literature survey is to identify information relevant to project work and the potential and known impacts of it within the project area. This section should include a comprehensive report of current market survey done with respect to problem. Include study of similar systems available, if any along with their pros and cons. identify those areas where there is an absence or scarcity.

II) Analysis

- Project plan
- Requirement analysis
- Team structure

The Group is to submit a detailed write-up indicating the requirements that the project demands-

Actual detailed problem definition

The definition is to include all that is to be done and is to be developed in the final software and / or Hardware (product) that is to be generated from the years' work (User's point of view).

The entries under this section are to be categorized under the categories,

1. Necessary functions,
2. Desirable functions,
3. Others

Requirement may not be final and provision should be available to add features dynamically without affecting the actual flow and design of the document. Modified Requirements (after doing feasibility study) are to be prepared under all the 3 categories listed above from the developer's point of view. The requirements listed herein should be feasible technically from the Software / Hardware point of view. The new list is also be categorized in the 3 categories listed above. (Follow the IEEE format of SRS)

The group is to suggest the Paradigm followed by them in the Project. The Paradigm should be justifiable from Phase I. The various stages and work to be completed under them is to be indicted in detail.

III) Design

- Software Requirement Specification(SRS) format is as given below.
- Risk assessment
- Brief discussion on Project plan submitted in semester I including major

milestones

and the work done as per it.

IV) Modelling

- UML diagrams (all 9)
- ERD & Normalization (NF) for database (if any)

ERDs (Optional, decide in consultation with guide)

- The group is to draw the ERD (Entity Relationship Diagram) for the Project. (This should be justifiable with regard to Phase I & II)
- The ERD after getting evaluated (by dry running) is to be analyzed for incompleteness from any point of view
- The ERD thus validated should be made fair in a presentable fashion
- This ERD is to be included in the Report

IF

The project group is to follow an "Object Oriented "Approach for their Project.

THEN

- The group is to draw all (all the nine) UML (Unified Modeling Language) diagrams for the project
- These diagrams are to be refined in every aspect for this report (as per requirements finalized in phase I)
- Proper notations are to be used in all the figures drawn
- Proper Color-coding if required is to be used

- Extensions to diagrams / customizations may be done and represented (if the project demands it)

ELSE (groups following Structured Approach)

- The group is to draw the DFD-s (Data Flow Diagrams) for the Project. (These should be justifiable with respect to Phase I, II and the ERD)
- DFD Level 0, Level 1, Level 2 should be drawn in an evolutionary fashion (No entries to appear in Level 2 unless they are in Level 1, which in turn are in Level 0)
- The DFD-s are to be validated and made final in a presentable fashion.
- Proper notations are to be used in all the figures drawn
- Proper Color-coding is expected
- Extensions to DFD-s may be represented (if the project demands it)

V) Coding

- Algorithms/ Flowcharts
- Software used
- Hardware specification
- Programming language
- Platform
- Components
- Tools
- Coding Style Format

V) Result Sets

VI) Testing

- Format technical reviews
 - Test plan
 - Test cases
 - Test results
- (Unit, integration, regression, system, α , β)

VII) Configuration Management Plan

VIII) Software Quality Assurance Plan

- * Costing (Time, Money and Resources)
- (Do not include costing in the project report; submit to the guide)

Conclusion

Your conclusions from the topic dealt with a minimum of 50 words.

References

List out Books, Magazines, Thesis, Journals, Web links etc referred in IEEE format

Glossary

In Keyword Alphabetical Order Ascending along with Page numbers

Plagiarism Check Report

Annexure iv: Project Report Formatting Guidelines

- 1) **Report Size:** Limit your Project report to preferably 25- 40 pages for partial project report. Limit your Project report to preferably 80-100 pages for final project report.
- 2) **Footer:** The footer “Department of Computer Engineering, SKNCOE Pune.” should be included. It should be TIMES NEW ROMAN 10 pt. and centrally justified.
- 3) **Header:** Project Title centered and page nos. on right should be included. **Start numbering from introduction.**
- 4) **Paper Size:** A4 Size, bond paper.
- 5) **Margins: Mirrored.**
 1. **Top** : 1 inch
 2. **Bottom** : 1 inch
 3. **Inside** :1.25 inch
 4. **Outside** :1 inch
- 6) **Line Spacing:** 1. 5 lines
- 7) **Title of Chapter:**
 1. **Font** : Arial (Bold face, Capital,)
 2. **Size** :16 pt. **Alignment:** centered
- 8) **All Topics heading:**
 - i. **First order Heading: (for example -1. Introduction)**
 1. **Font** : Times New Roman(Bold Face)
 2. **Size** : 14 pt.
 - ii. **Second order Heading: (for example -1.1 Evolution)**
 1. **Font** : Times New Roman(Bold Face)
 2. **Size** : 12 pt.
 - iii. **Third order Heading: for example -1.1.1 Image Processing**
 1. **Font** : Times New Roman(Normal Face)
 2. **Size** : 12 pt.
- 9) **Text:**
 1. **Font** : Times New Roman(Bold Face)
 2. **Size** : 12 pt
- 10) **Figures and Tables:**
 1. **Caption:**(for figures below the figure and for tables above the table)
 2. **Font:** Garamond(Bold)
 3. **Size:**11 pt.
 4. **Alignment:** Center
- 11) **References:**
 - . **Book**
Author name(s), Book Title, Publisher, Copyright Year, page nos. if any.
 - . **Journal/ Magazine/ Periodical**
Author name(s), paper name, Journal/ Magazine/ Periodical name, issue no., page nos.
 - . **Web Resources**
Complete URL including File name.

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Annexure vii: Report Documentation

Report Documentation				
Report Code: CS-BE-Project 2025-2026			Report Number:	
Report Title: LegalSetu - Legal Compliance & Lawyer Consultation Portal				
Address (Details): Parvatibai Genba Moze College of Engineering Wagholi, Gate No -2181, Pune-412207, India.				
Author 1: Adil Shaikh [with Phone: 781096206 Address E-mail : adilofficial@gmail.com Roll: Cell No	Author 2 : Shrikant Gavhale [with Phone : 7841920798 Address E-mail : Shrikantgavhale151@gmail.com Roll: Cell No	Author 3 : Mohit Badgujar [with Phone 9370949271, Address E-mail : mohitbadgujar16@gmail.com Roll: Cell No		
Year: 2025-2026 Branch: Computer Engineering				
Key Words: <Keywords in the Report>				
Type of Report: FINAL	Report Checked By:	Report Checked Date:	Guides Complete Name:	Total Copies

Abstract: <A Brief Abstract of the Seminar>

NOTE –

This table should not go beyond this page.

Scale down the Abstract if it does not fit in one page.

Take guide's Signature in the **“Report Checked by:”** Cell and Date of Signature in the **“Report Checked Date:”** Cell.

This page is the last page of the projects report and is NOT to be included in the **“Page Count”**